

Chapter 14

Incomplete Block Designs

Block designs group similar units into blocks so that variation among units within the blocks is reduced. Complete block designs, such as RCB and LS, have each treatment occurring once in each block. Incomplete block designs also group units into blocks, but the blocks do not have enough units to accommodate all the treatments.

Not all treatments
appear in an
incomplete block

Incomplete block designs share with complete block designs the advantage of variance reduction due to blocking. The drawback of incomplete block designs is that they do not provide as much information per experimental unit as a complete block design with the same error variance. Thus complete blocks are preferred over incomplete blocks when both can be constructed with the same error variance.

Incomplete blocks
less efficient than
complete blocks

Eyedrops

Eye irritation can be reduced with eyedrops, and we wish to compare three brands of eyedrops for their ability to reduce eye irritation. (There are problems here related to measuring eye irritation, but we set them aside for now.) We expect considerable subject to subject variation, so blocking on subject seems appropriate. If each subject can only be used during one treatment period, then we must use one brand of drop in the left eye and another brand in the right eye. We are forced into incomplete blocks of size two, because our subjects have only two eyes.

Suppose that we have three subjects that receive brands (A and B), (A and C), and (B and C) respectively. How can we estimate the expected difference in responses between two treatments, say A and B? We can get some information from subject 1 by taking the difference of the A and B responses; the

Example 14.1

subject effect will cancel in this difference. This first difference has variance $2\sigma^2$. We can also get an estimate of A-B by subtracting the B-C difference in subject three from the A-C difference in subject two. Again, subject effects cancel out, and this difference has variance $4\sigma^2$. Similar approaches yield estimates of A-C and B-C using data from all subjects.

If we had had two complete blocks (three-eyed subjects?) with the same unit variance, then we would have had two independent estimates of A-B each with variance $2\sigma^2$. Thus the incomplete block design has more variance in its estimates of treatment differences than does the complete block design with the same variance and number of units.

There are many kinds of incomplete block designs. This chapter will cover only some of the more common types. Several of the incomplete block designs given in this chapter have “balanced” in their name. It is important to realize that these designs are not balanced in the sense that all block and factor-level combinations occur equally often. Rather they are balanced using somewhat looser criteria that will be described later.

Resolvable
designs split into
replications

Two general classes of incomplete block designs are *resolvable* designs and *connected* designs. Suppose that each treatment is used r times in the design. A resolvable design is one in which the blocks can be arranged into r groups, with each group representing a complete set of treatments. Resolvable designs can make management of experiments simpler, because each replication can be run at a different time or a different location, or entire replications can be dropped if the need arises. The eyedrop example is not resolvable.

Connected
designs can
estimate all
treatment
differences

A design is *disconnected* if you can separate the treatments into two groups, with no treatment from the first group ever appearing in the same block with a treatment from the second group. A *connected* design is one that is not disconnected. In a connected design you can estimate all treatment differences. You cannot estimate all treatment differences in a disconnected design; in particular, you cannot estimate differences between treatments in different groups. Connectedness is obviously a very desirable property.

14.1 Balanced Incomplete Block Designs

BIBD

The Balanced Incomplete Block Design (BIBD) is the simplest incomplete block design. We have g treatments, and each block has k units, with $k < g$. Each treatment will be given to r units, and we will use b blocks. The total number of units N must satisfy $N = kb = rg$. The final requirement for a BIBD is that all pairs of treatments must occur together in the same number of blocks. The BIBD is called “balanced” because the variance of the estimated

Table 14.1: Plates washed before foam disappears. Letters indicate treatments.

Session											
1	2	3	4	5	6	7	8	9	10	11	12
A 19	D 6	G 21	A 20	B 17	C 15	A 20	B 16	C 13	A 20	B 17	C 14
B 17	E 26	H 19	D 7	E 26	F 23	E 26	F 23	D 7	F 24	D 6	E 24
C 11	F 23	J 28	G 20	H 19	J 31	J 31	G 21	H 20	H 19	J 29	G 21

difference of treatment effects $\hat{\alpha}_i - \hat{\alpha}_j$ is the same for all pairs of treatments i, j .

Example 14.1 is the simplest possible BIBD. There are $g = 3$ treatments, with blocks of size $k = 2$. Each treatment occurs $r = 2$ times in the $b = 3$ blocks. There are $N = 6$ total units, and each pair of treatments occurs together in one block.

We may use the BIBD design for treatments with factorial structure. For example, suppose that we have three factors each with two levels for a total of $g = 8$ treatments. If we have $b = 8$ blocks of size $k = 7$, then we can use a BIBD with $r = 7$, with each treatment left out of one block and each pair of treatments occurring together six times.

Dish detergent

John (1961) gives an example of a BIBD. Nine different dishwashing solutions are to be compared. The first four consist of base detergent I and 3, 2, 1, and 0 parts of an additive; solutions five through eight consist of base detergent II and 3, 2, 1, and 0 parts of an additive; the last solution is a control. There are three washing basins and one operator for each basin. The three operators wash at the same speed during each test, and the response is the number of plates washed when the foam disappears. The speed of washing is the same for all three detergents used at any one session, but could differ from session to session.

Table 14.1 gives the design and the results. There are $g = 9$ treatments arranged in $b = 12$ incomplete blocks of size $k = 3$. Each treatment appears $r = 4$ times, and each pair of treatments appears together in one block.

The requirement that all pairs of treatments occur together in an equal number of blocks is a real stickler. Any given treatment occurs in r blocks, and there are $k - 1$ other units in each of these blocks for a total of $r(k - 1)$ units. These must be divided evenly between the $g - 1$ other treatments. Thus $\lambda = r(k - 1)/(g - 1)$ must be a whole number for a BIBD to exist. For the eyedrop example, $\lambda = 2(2 - 1)/(3 - 1) = 1$, and for the dishes example, $\lambda = 4(3 - 1)/(9 - 1) = 1$.

Example 14.2

Treatment pairs
occur together λ
times

A major impediment to the use of the BIBD is that no BIBD may exist for your combination of $kb = rg$. For example, you may have $g = 5$ treatments and $b = 5$ blocks of size $k = 3$. Then $r = 3$, but $\lambda = 3(3-1)/(5-1) = 3/2$ is not a whole number, so there can be no BIBD for this combination of r , k , and g . Unfortunately, λ being a whole number is not sufficient to guarantee that a BIBD exists, though one usually does.

Unreduced BIBD
has all
combinations

BIBD tables

A BIBD always exists for every combination of $k < g$. For example, you can always generate a BIBD by using all combinations of the g treatments taken k at a time. Such a BIBD is called *unreduced*. The problem with this approach is that you may need a lot of blocks for the design. For example, the unreduced design for $g = 8$ treatments in blocks of size $k = 4$ requires $b = 70$ blocks. Appendix C contains a list of some BIBD plans for $g \leq 9$. Fisher and Yates (1963) and Cochran and Cox (1957) contain much more extensive lists.

Design
complement

Symmetric BIBD

If you have a plan for a BIBD with g , k , and b blocks, then you can construct a plan for g treatments in b blocks of $g - k$ units per block simply by using in each block of the second design the treatments *not* used in the corresponding block of the first design. The second design is called the *complement* of the first design. When $b = g$ and $r = k$, a BIBD is said to be *symmetric*. The eyedrop example above is symmetric; the detergent example is not symmetric.

BIBD
randomization

Randomization of a BIBD occurs in three steps. First, randomize the assignment of physical blocks to subgroups of treatment letters (or numbers) given in the design. Second, randomize the assignment of these treatment letters to physical units within blocks. Third, randomize the assignment of treatment letters to treatments.

14.1.1 Intrablock analysis of the BIBD

BIBD model

Intrablock analysis sounds exotic, but it is just the standard analysis that you would probably have guessed was appropriate. Let y_{ij} be the response for treatment i in block j ; we do not observe all i, j combinations. Use the model

$$y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij} \quad .$$

If treatments are fixed, we assume that the treatment effects sum to zero; otherwise we assume that they are a random sample from a $N(0, \sigma_\alpha^2)$ distribution. Block effects may be fixed or random.

Our usual methods for estimating treatment effects *do not* work for the BIBD. In this way, this “balanced” design is more like an unbalanced factorial or an RCB with missing data. For those situations, we relied on statistical

Listing 14.1: SAS output for intrablock analysis of detergent data.

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	19	1499.56481	78.92446	95.77	0.0001
Error	16	13.18519	0.82407		①
Source	DF	Type III SS	Mean Square	F Value	Pr > F
BLOCK	11	10.06481	0.91498	1.11	0.4127
DETERG	8	1086.81481	135.85185	164.85	0.0001 ②
Contrast	DF	Contrast SS	Mean Square	F Value	Pr > F
control vs test	1	345.041667	345.041667	418.70	0.0001 ③
base I vs base II	1	381.337963	381.337963	462.75	0.0001
linear in additive	1	306.134259	306.134259	371.49	0.0001
Parameter	Estimate	T for H0: Parameter=0	Pr > T	Std Error of Estimate	
base I vs base II	-7.9722222	-21.51	0.0001	0.37060178	④

software to fit the model, and we do so here as well. Similarly, our usual contrast methods do not work either. An RCB with missing data is a good way to think about the analysis of the BIBD, even though in the BIBD the data were planned to be missing in a very systematic way.

For the RCB with missing data, we computed the sum of squares for treatments adjusted for blocks. That is, we let blocks account for as much variation in the data as they could, and then we determined how much additional variation could be explained by adding treatments to the model. Because we had already removed the variation between blocks, this additional variation explained by treatments must be variation within blocks: hence *in-trablock* analysis. Intrablock analysis of a BIBD is analysis with treatments adjusted for blocks.

Usual estimates of treatment effects do not work for BIBD

Intrablock analysis is treatments adjusted for blocks

Dish detergent, continued

The basic intrablock ANOVA consists of treatments adjusted for blocks. Listing 14.1 ② shows SAS output for this model; the Type III sum of squares for detergent is adjusted for blocks. Residual plots show that the variance is fairly stable, but the residuals have somewhat short tails. There is strong evidence against the null hypothesis (p -value .0001).

Example 14.3

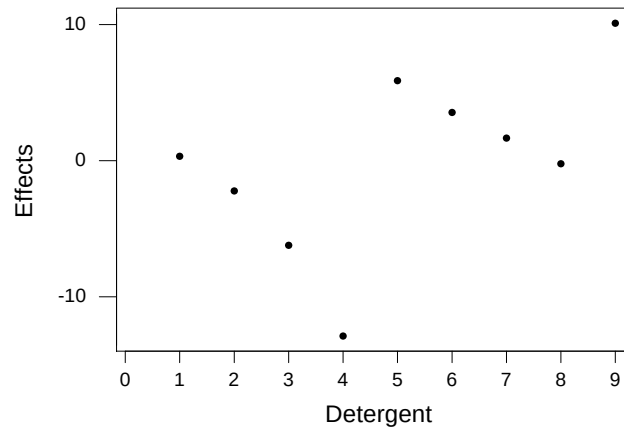


Figure 14.1: Treatment effects for intrablock analysis of dish detergent data, using Minitab.

We can examine the treatment effects more closely by comparing the two detergent bases with each other and the control, and by looking at the effects of the additive. Figure 14.1 shows the nine treatment effects. Clearly there is a mostly linear effect due to the amount of additive, with more additive giving a higher response. We also see that detergent base I gives lower responses than detergent base II, and both are lower than the control. For example, the contrast between base I and base II has sum of squares 381.34; the contrast between the control and the other treatments has sum of squares 345.04; and the linear in additive contrast has sum of squares 306.16 (Listing 14.1 ③). These 3 degrees of freedom account for 1032.5 of the total 1086.8 sum of squares between treatments.

Efficiency of BIBD
to RCB

There is in fact a fairly simple hand-calculation for treatments adjusted for blocks in the BIBD; the availability of this simple calculation helped make the BIBD attractive before computers. We discuss the calculation not because you will ever be doing the calculations that way, but rather because it helps give some insight into $E_{\text{BIBD:RCB}}$, the efficiency of the BIBD relative to the RCB. Define $E_{\text{BIBD:RCB}}$ to be

$$E_{\text{BIBD:RCB}} = \frac{g(k-1)}{(g-1)k} ,$$

where g is the number of treatments and k is the number of units per block. Observe that $E_{\text{BIBD:RCB}} < 1$, because $k < g$ in the BIBD. For the detergent example, $E_{\text{BIBD:RCB}} = 9 \times 2 / (8 \times 3) = 3/4$.

The value $E_{\text{BIBD:RCB}}$ is the relative efficiency of the BIBD to an RCB with the same variance. One way to think about $E_{\text{BIBD:RCB}}$ is that every unit in a BIBD is only worth $E_{\text{BIBD:RCB}}$ units worth of information in an RCB with the same variance. Thus while each treatment is used r times in a BIBD, the effective sample size is only $rE_{\text{BIBD:RCB}}$.

Effective sample
size $rE_{\text{BIBD:RCB}}$

The hand-calculation formulae for the BIBD use the effective sample size in place of the actual sample size. Let $\bar{y}_{\bullet j}$ be the mean response in the j th block; let $v_{ij} = y_{ij} - \bar{y}_{\bullet j}$ be the data with block means removed; and let $v_{i\bullet}$ be the sum of the v_{ij} values for treatment i (there are r of them). Then we have

Hand formulae for
BIBD use
effective sample
size

$$\hat{\alpha}_i = \frac{v_{i\bullet}}{rE_{\text{BIBD:RCB}}} ,$$

$$SS_{\text{Tt}} = \sum_{i=1}^g (rE_{\text{BIBD:RCB}}) \hat{\alpha}_i^2 ,$$

and

$$\text{Var}\left(\sum_i w_i \hat{\alpha}_i\right) = \sigma^2 \sum_i \frac{w_i^2}{rE_{\text{BIBD:RCB}}} .$$

We can also use pairwise comparison procedures with the effective sample size.

In practice, we can often find incomplete blocks with a smaller variance σ_{bibd}^2 than can be attained using complete blocks σ_{rcb}^2 . We prefer the BIBD design over the RCB if

BIBD beats RCB
if variance
reduction great
enough

$$\frac{\sigma_{\text{bibd}}^2}{rE_{\text{BIBD:RCB}}} < \frac{\sigma_{\text{rcb}}^2}{r}$$

or

$$\frac{\sigma_{\text{bibd}}^2}{\sigma_{\text{rcb}}^2} < E_{\text{BIBD:RCB}} ;$$

in words, we prefer the BIBD if the reduction in variance more than compensates for the loss of efficiency. This comparison ignores adjustments for error degrees of freedom.

14.1.2 Interblock information

Recovery of
interblock
information when
block effects are
random

The first thing we did in the intrablock analysis of the BIBD was to subtract block means from the data to get deviations from the block means. When the block effects in a BIBD are random effects, then these block means also contain information about treatment differences. We can use block means or block totals to produce a second set of estimates for treatment effects, called the *interblock* estimates, independent of the usual intrablock estimates. Combining the interblock and intrablock estimates is called “recovery of interblock information.”

First get
intrablock
estimate

Suppose that we want to estimate a contrast $\zeta = \sum_i w_i \alpha_i$. Recovery of interblock information takes place in three steps. First, compute the *intrablock* estimate of the contrast and its variance. Second, compute the *interblock* estimate of the contrast and its variance. Third, combine the two estimates. The intrablock estimate is simply the standard estimate of the last section:

$$\hat{\zeta} = \sum_{i=1}^g w_i \hat{\alpha}_i$$

with variance

$$Var(\hat{\zeta}) = \sigma^2 \sum_{i=1}^g \frac{w_i^2}{r E_{\text{BIBD:RCB}}} ,$$

using MS_E to estimate σ^2 .

For step 2, start by letting n_{ij} be 1 if treatment i occurs in block j , and 0 otherwise. Then the block total $y_{\bullet j}$ can be expressed

$$\begin{aligned} y_{\bullet j} &= k\mu + \sum_{i=1}^g n_{ij} \alpha_i + \left\{ k\beta_j + \sum_{i=1}^g n_{ij} \epsilon_{ij} \right\} \\ &= k\mu + \sum_{i=1}^g n_{ij} \alpha_i + \eta_j . \end{aligned}$$

Interblock
estimates from
block totals

This has the form of a multiple regression with g predicting variables and an independent and normally distributed error η_j having variance $k^2 \sigma_\beta^2 + k\sigma^2$. Some tedious algebra shows that the interblock estimates are

$$\begin{aligned} \tilde{\mu} &= \bar{y}_{\bullet\bullet} \\ \tilde{\alpha}_i &= \frac{\sum_{j=1}^b n_{ij} y_{\bullet j} - r k \tilde{\mu}}{r - \lambda} , \end{aligned}$$

and the variance of the contrast $\tilde{\zeta} = \sum_{i=1}^g w_i \tilde{\alpha}_i$ is

$$Var(\tilde{\zeta}) = (k^2 \sigma_\beta^2 + k \sigma^2) \sum_{i=1}^g \frac{w_i^2}{r - \lambda} .$$

We estimate σ^2 using the MS_E from the intrablock analysis. Estimating σ_β^2 involves something highly unusual. The expected value of the mean square for *blocks adjusted for treatments* is $\sigma^2 + (N - g)\sigma_\beta^2/(b - 1)$. Thus an unbiased estimate of σ_β^2 is

$$\hat{\sigma}_\beta^2 = \frac{b - 1}{N - g} (MS_{\text{blocks adjusted}} - MS_E) .$$

Use blocks adjusted for treatments to get block variance

This interblock recovery is the only place we will consider blocks adjusted for treatments.

At this stage, we have the intrablock estimate $\hat{\zeta}$ and its variance $Var(\hat{\zeta})$, and we have the interblock estimate $\tilde{\zeta}$ and its variance $Var(\tilde{\zeta})$. If the variances were equal, we would just average the two estimates to get a combined estimate. However, the variance of the intrablock estimate is always less than the interblock estimate, so we want to give the intrablock estimate more weight in the average. The best weight is “inversely proportional to the variance”, so the combined estimate for contrast ζ is

$$\bar{\zeta} = \frac{\frac{1}{Var(\hat{\zeta})} \hat{\zeta} + \frac{1}{Var(\tilde{\zeta})} \tilde{\zeta}}{\frac{1}{Var(\hat{\zeta})} + \frac{1}{Var(\tilde{\zeta})}} .$$

Use weighted average to combine inter- and intrablock estimates

This combined estimate has variance

$$Var(\bar{\zeta}) = \frac{1}{\frac{1}{Var(\hat{\zeta})} + \frac{1}{Var(\tilde{\zeta})}} .$$

Dish detergent, continued

Suppose that we wish to examine the difference between detergent bases I and II. We can do that with a contrast w with coefficients (.25, .25, .25,

Example 14.4

.25, -.25, -.25, -.25, -.25, 0). Listing 14.1 ④ shows that this contrast has an estimated value of -7.972 with a standard error of .3706 (variance .1373); this is the intrablock estimate.

We begin the interblock analysis by getting the block totals, the incidence matrix $\{n_{ij}\}$ (shown here with treatments indexing columns), and the sums of the cross products:

Block total	Treatment incidence								
	1	2	3	4	5	6	7	8	9
47	1	1	1	0	0	0	0	0	0
55	0	0	0	1	1	1	0	0	0
68	0	0	0	0	0	0	1	1	1
47	1	0	0	1	0	0	1	0	0
62	0	1	0	0	1	0	0	1	0
69	0	0	1	0	0	1	0	0	1
77	1	0	0	0	1	0	0	0	1
60	0	1	0	0	0	1	1	0	0
40	0	0	1	1	0	0	0	1	0
63	1	0	0	0	0	1	0	1	0
52	0	1	0	1	0	0	0	0	1
59	0	0	1	0	1	0	1	0	0
$\sum_j n_{ij}\bar{y}_{\bullet j}$	234	221	215	194	253	247	234	233	266

Applying the formula, we get that the interblock estimates are .333, -4, -6, -13, 6.667, 4.667, .333, 0, and 11. The interblock estimate $\tilde{\zeta}$ is thus

$$\tilde{\zeta} = (.333 - 4 - 6 - 13)/4 - (6.667 + 4.667 + .333 + 0)/4 = -8.583 .$$

The variance of $\tilde{\zeta}$ is

$$\begin{aligned} Var(\tilde{\zeta}) &= (k^2\sigma_\beta^2 + k\sigma^2) \sum_{i=1}^a \frac{w_i^2}{r - \lambda} \\ &= (9\sigma_\beta^2 + 3\sigma^2) \frac{8 \times .25^2}{3} \\ &= (3\sigma_\beta^2 + \sigma^2)/2 \end{aligned}$$

The intrablock MS_E of .82407 estimates σ^2 (Listing 14.1 ①). The mean square for blocks adjusted for treatments is .91498 from Listing 14.1 ②. (We

show Type III sums of squares, so blocks are also adjusted for treatments.) The estimate for σ_β^2 is thus

$$\begin{aligned}\hat{\sigma}_\beta^2 &= \frac{b-1}{N-g}(MS_{\text{blocks adjusted}} - MS_E) \\ &= \frac{11}{27}(.91498 - .82407) \\ &= .0370\end{aligned}$$

Substituting in, we get

$$\begin{aligned}Var(\tilde{\zeta}) &= (3\sigma_\beta^2 + \sigma^2)/2 \\ &= (3 \times .0370 + .82407)/2 \\ &= .4675\end{aligned}$$

Note that even with an estimated block variance of nearly zero, the intrablock estimate of the contrast is still much more precise than the interblock estimate.

The intrablock estimate and variance are -7.972 and .1374, and the interblock estimate and variance are -8.583 and .4675. The combined estimate is

$$\begin{aligned}\bar{\zeta} &= \frac{\frac{-7.972}{.1374} + \frac{-8.583}{.4675}}{\frac{1}{.1374} + \frac{1}{.4675}} \\ &= -8.111\end{aligned}$$

with variance

$$\begin{aligned}Var(\bar{\zeta}) &= \frac{1}{\frac{1}{.1374} + \frac{1}{.4675}} \\ &= .1062\end{aligned}$$

That was a lot of work. Unfortunately, this effort often provides minimal improvement over the intrablock estimates. When there is no block variance (that is, when $\sigma_\beta^2 = 0$), then the interblock variance for a contrast is $g(k-1)/(g-k)$ times as large as the intrablock variance. When blocking is

Interblock recovery often provides little improvement

successful, the variation between blocks will be large compared to the variation within blocks. Then the variance of intrablock estimates will be much smaller than those of interblock estimates, and the combined estimates are very close to the intrablock estimates.

Weights are variable

Another fact to bear in mind is that the weights used in the weighted average to combine intra- and interblock information are rather variable when b is small. This variation comes from the ratio $MS_{\text{blocks adjusted}}/MS_E$, which appears in the formula for the weights. As we saw when trying to estimate ratios of variance components, we need quite a few degrees of freedom in both the numerator and denominator before the ratio, and thus the weights, are stable.

14.2 Row and Column Incomplete Blocks

We use Latin Squares and their variants when we need to block on two sources of variation in complete blocks. We can use *Youden Squares* when we need to block on two sources of variation, but cannot set up the complete blocks for LS designs. I've always been amused by this name, because Youden Squares are not square.

Youden Squares are incomplete Latin Squares

The simplest example of a Youden Square starts with a Latin Square and deletes one of the rows (or columns). The resulting arrangement has g columns and $g - 1$ rows. Each row is a complete block for the treatments, and the columns form an unreduced BIBD for the treatments. Here is a simple Youden Square formed from a four by four Latin Square:

A	B	C	D
B	A	D	C
C	D	A	B

Youden Square is BIBD on columns and RCB on rows

A more general definition of a Youden Square is a rectangular arrangement of treatments, with the columns forming a BIBD and all treatments occurring an equal number of times in each row. In particular, any symmetric BIBD ($b = g$) can be rearranged into a Youden Square. For example, here is a symmetric BIBD with $g = b = 7$ and $r = k = 3$ arranged as a Youden Square:

A	B	C	D	E	F	G
B	C	D	E	F	G	A
D	E	F	G	A	B	C

Table 14.2: Serum levels of lithium ($\mu\text{Eq/l}$) 12 hours after administration. Treatments are 300 mg and 250 mg capsules, 450 mg time delay capsule, and 300 mg solution.

Week	Subject					
1	A 200	D 267	C 156	B 280	D 333	D 233
2	B 160	C 178	A 200	C 178	A 167	B 200
1	B 320	B 320	C 111	A 333	A 233	C 244
2	A 200	D 200	D 133	D 200	C 178	B 160

In Appendix C, those BIBD's that can be arranged as Youden Squares are so arranged.

The analysis of a Youden Square is a combination of the Latin Square and BIBD, as might be expected. Because both treatments and columns appear once in each row, row contrasts are orthogonal to treatment and column contrasts, and this makes computation a little easier. Youden Squares are also called *row orthogonal* for this reason. The intrablock ANOVA has terms for rows, columns, treatments (adjusted for columns), and error. Row effects and sums of squares are computed via the standard formulae, ignoring columns and treatments. Column sums of squares (unadjusted) are computed ignoring rows and treatments. Intrablock treatment effects and sums of squares are computed as for a BIBD with columns as blocks. Error sums of squares are found by subtraction. Interblock analysis of the Youden Square and the combination of inter- and intrablock information are exactly like the BIBD.

Row orthogonal
designs

Intrablock
analysis adjusts
for rows and
columns

Interblock
analysis similar to
BIBD

Lithium in blood

We wish to compare the blood concentrations of lithium 12 hours after administering lithium carbonate, using either a 300 mg capsule, 250 mg capsule, 450 mg time delay capsule, or 300 mg solution. There are twelve subjects, each of whom will be used twice, 1 week apart. We anticipate that the responses will be different in the second week, so we block on subject and week. The response is the serum lithium level as shown in Table 14.2 (data from Westlake 1974).

There are $g = 4$ treatments in $b = 12$ blocks of size $k = 2$, so that $r = 6$. We have $\lambda = 2$, $E = 2/3$, and each treatment appears three times in each week for a Youden Square.

The intrablock ANOVA for these data is shown in Listing 14.2. The residual plots (not shown) are passable. There is no evidence for a difference between the treatments 12 hours after administration. However, note that the

Example 14.5

Listing 14.2: Minitab output for intrablock analysis of lithium data.

Source	DF	Seq SS	Adj SS	Seq MS	F	P
week	1	0.031974	0.031974	0.031974	15.79	0.004
subject	11	0.039344	0.029946	0.003577	1.77	0.215
treatmen	3	0.005603	0.005603	0.001868	0.92	0.473
Error	8	0.016203	0.016203	0.002025		

mean square for the week blocking factor is fairly large. If we had ignored the week effect, we could anticipate an error mean square of

$$\frac{11 \times .0020253 + .031974}{12} = .00452 ,$$

more than doubling the error mean square in the Youden Square design.

14.3 Partially Balanced Incomplete Blocks

BIBD's are too big
for some g and k

BIBD's are great, but their balancing requirements may imply that the smallest possible BIBD for a given g and k is too big to be practical. For example, let's look for a BIBD for $g = 12$ treatments in incomplete blocks of size $k = 7$. To be a BIBD, $\lambda = r(k - 1)/(g - 1) = 6r/11$ must be a whole number; this implies that r is some multiple of 11. In addition, $b = rg/k = (11 \times m) \times 12/7$ must be a whole number, and that implies that b is a multiple of $11 \times 12 = 132$. So the smallest possible BIBD has $r = 77$, $b = 132$, and $N = 924$. This is a bigger experiment that we are likely to run.

PBIBD has
 $N = gr = bk$;
some treatment
pairs more
frequent

Partially Balanced Incomplete Block Designs (PBIBD) allow us to run incomplete block designs with fewer blocks than may be required for a BIBD. The PBIBD has g treatments and b blocks of k units each; each treatment is used r times, and there is a total of $N = gr = bk$ units. The PBIBD does not have the requirement that each pair of treatments occurs together in the same number of blocks. This in turn implies that not all differences $\hat{\alpha}_i - \hat{\alpha}_j$ have the same variance in a PBIBD.

Sample PBIBD

Here is a sample PBIBD with $g = 12$, $k = 7$, $r = 7$, and $b = 12$. In this representation, each row is a block, and the numbers in the row indicate which treatments occur in that block.

Block	Treatments						
1	1	2	3	4	5	8	10
2	2	3	4	5	6	9	11
3	3	4	5	6	7	10	12
4	1	4	5	6	7	8	11
5	2	5	6	7	8	9	12
6	1	3	6	7	8	9	10
7	2	4	7	8	9	10	11
8	3	5	8	9	10	11	12
9	1	4	6	9	10	11	12
10	1	2	5	7	10	11	12
11	1	2	3	6	8	11	12
12	1	2	3	4	7	9	12

We see, for example, that treatment 1 occurs three times with treatments 5 and 9, and four times with all other treatments.

The design rules for a PBIBD are fairly complicated:

1. There are g treatments, each used r times. There are b blocks of size $k < g$. Of course, $bk = gr$. No treatment occurs more than once in a block.
2. There are m *associate classes*. Any pair of treatments that are i th associates appears together in λ_i blocks. We usually arrange the λ_i values in decreasing order, so that first associates appear together most frequently.
3. All treatments have the same number of i th associates, namely ρ_i .
4. Let A and B be two treatments that are i th associates, and let p_{jk}^i be the number of treatments that are j th associates of A and k th associates of B. This number p_{jk}^i does not depend on the pair of i th associates chosen. In particular, $p_{jk}^i = p_{kj}^i$.

Requirements for
PBIBD

Associate classes

ρ_i i th associates

The PBIBD is partially balanced, because the variance of $\hat{\alpha}_i - \hat{\alpha}_j$ depends upon whether i, j are first, second, or m th associates. The randomization of a PBIBD is just like that for a BIBD.

Randomize
PBIBD like BIBD

Let's check the design given above and verify that it is a PBIBD. First note that $g = 12$, $k = 7$, $r = 7$, $b = 12$, and no treatment appears twice in a block. Next, there are two associate classes, with first associates appearing together four times and second associates appearing together three times. The pairs (1,5), (1,9), (2,6), (2,10), (3,7), (3,11), (4,8), (4,12), (5,9), (6,10), (7,11), and (8,12) are second associates; all other pairs are first associates. Each treatment has nine first associates and two second associates. For any pair of

first associates, there are six other treatments that are first associates of both, four other treatments that are first associates of one and second associates of the other (two each way), and no treatments that are second associates of both. We thus have

$$\{p_{ij}^1\} = \begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix}.$$

For any pair of second associates, there are nine treatments that are first associates of both, and one treatment that is a second associate of both, so that

$$\{p_{ij}^2\} = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus all the design requirements are met, and the example design is a PBIBD.

Intrablock
analysis is
treatments
adjusted for
blocks

One historical advantage of the PBIBD was that the analysis could be done by hand. That is, there are relatively simple expressions for the various intra- and interblock analyses. With computers, that particular advantage is no longer very useful. The intrablock analysis of the PBIBD is simply treatments adjusted for blocks, as with the BIBD.

PBIBD less
efficient on
average than
BIBD

The efficiency of a PBIBD is actually an average efficiency. The variance of $\hat{\alpha}_i - \hat{\alpha}_j$ depends on whether treatments i and j are first associates, second associates, or whatever. So to compute efficiency $E_{\text{PBIBD:RCB}}$, we divide the variance obtained in an RCB for a pairwise difference ($2\sigma^2/r$) by the average of the variances of all pairwise differences in the PBIBD. There is an algorithm to determine $E_{\text{PBIBD:RCB}}$, but there is no simple formula. We can say that the efficiency will be less than $g(k-1)/[(g-1)k]$, which is the efficiency of a BIBD with the same block size and number of treatments.

There are several extensive catalogues of PBIBD's, including Bose, Clatworthy, and Shrikhande (1954) (376 separate designs) and Clatworthy (1973).

14.4 Cyclic Designs

Cyclic designs
are simple

Cyclic designs are easily constructed incomplete block designs that permit the study of g treatments in blocks of size k . We will only examine the simplest situation, where the replication r for each treatment is a multiple of k , the block size. So $r = mk$, and $b = mg$ is the number of blocks. Cyclic designs include some BIBD and PBIBD designs.

Cycles of
treatments

A cycle of treatments starts with an initial treatment and then proceeds through the subsequent treatments in order. Once we get to treatment g , we go back down to treatment 1 and start increasing again. For example, with seven treatments we might have the cycle (4, 5, 6, 7, 1, 2, 3).

Cyclic construction starts with an initial block and builds $g - 1$ more blocks from the initial block by replacing each treatment in the initial block by its successor in the cycle. Additional sets of g blocks are constructed from new initial blocks. Thus all we need to know to build the design are the initial blocks.

Proceed through
cycles from initial
block

Write the initial block in a column, and write the cycles for each treatment in the initial block in rows, obtaining a k by g arrangement. The columns of this arrangement are the blocks. For example, suppose we have seven treatments and the initial block [1,4]. The cyclic design has blocks (columns):

$$\begin{array}{c|c|c|c|c|c|c} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline 4 & 5 & 6 & 7 & 1 & 2 & 3 \end{array}$$

Each row is a cycle started by a treatment in the initial block. Cycles are easy, so cyclic designs are easy, once you have the initial block.

But wait, there's more! Not only do we have an incomplete block design with the columns as blocks, we have a complete block design with the rows as blocks. Thus cyclic designs are row orthogonal designs (and may be Youden Squares if the cyclic design is BIBD).

Cyclic designs
are row
orthogonal

Appendix C.3 contains a table of initial blocks for cyclic designs for k from 2 through 10 and g from 6 through 15. Several initial blocks are given for the smaller designs, depending on how many replications are required. For example, for $k = 3$ the table shows initial blocks for 3, 6, and 9 replications. Use the first initial block if $r = 3$, use the first and second initial blocks if $r = 6$, and use all three initial blocks if $r = 9$. For $g = 10$, $k = 3$, and $r = 6$, the initial blocks are (1,2,5) and (1,3,8), and the plan is

$$\begin{array}{c|c|c|c|c|c|c|c|c|c|c} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ \hline 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 1 \\ \hline 5 & 6 & 7 & 8 & 9 & 10 & 1 & 2 & 3 & 4 \end{array}$$

$$\begin{array}{c|c|c|c|c|c|c|c|c|c|c} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ \hline 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 1 & 2 \\ \hline 8 & 9 & 10 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}$$

As with the PBIBD, there is an algorithm to compute the (average) efficiency of a cyclic design, but there is no simple formula. The initial blocks given in Appendix C.3 were chosen to make the cyclic designs as efficient as possible.

14.5 Square, Cubic, and Rectangular Lattices

Lattice designs
for special g, k
combinations

Lattice designs work when the number of treatments g and the size of the blocks k follow special patterns. Specifically,

- A Square Lattice can be used when $g = k^2$.
- A Cubic Lattice can be used when $g = k^3$.
- A Rectangular Lattice can be used when $g = k(k + 1)$.

These lattice designs are resolvable and are most useful when we have a large number of treatments to be run in small blocks.

We illustrate the Square Lattice when $g = 9 = 3^2$. Arrange the nine treatments in a square; for example:

1	2	3
4	5	6
7	8	9

A simple lattice
has two
replications made
of rows and
columns of the
square

There is nothing special about this pattern; we could arrange the treatments in any way. The first replicate of the Square Lattice consists of blocks made up of the rows of the square: here (1, 2, 3), (4, 5, 6), and (7, 8, 9). The second replicate consists of blocks made from the columns of the square: (1, 4, 7), (2, 5, 8), and (3, 6, 9). A Square Lattice must have at least these two replicates to be connected, and a Square Lattice with only two replicates is called a *simple lattice*.

Triple lattice uses
Latin Square for
third replicate

We add a third replication using a Latin Square. A Square Lattice with three replicates is called a *triple lattice*. Here is a three by three Latin Square:

A	B	C
B	C	A
C	A	B

Assign treatments to blocks using the letter patterns from the square. The three blocks of the third replicate are (1, 6, 8), (2, 4, 9), and (3, 5, 7).

Additional
replicates use
orthogonal Latin
Squares

You can construct additional replicates for every Latin Square that is orthogonal to those already used. For example, the following square

A	B	C
C	A	B
B	C	A

is orthogonal to the first one used. Our fourth replicate is thus (1, 5, 9), (2, 6, 7), and (3, 4, 8). Recall that there are no six by six Graeco-Latin Squares (six by six orthogonal Latin Squares), so only simple and triple lattices are possible for $g = 6^2$.

For $g = k^2$, there are at most $k - 1$ orthogonal Latin Squares. The Square Lattice formed when $k - 1$ Latin Squares are used has $k + 1$ replicates; is called a *balanced lattice*; and is a BIBD with $g = k^2$, $b = k(k + 1)$, $r = k + 1$, $\lambda = 1$, and $E = k/(k + 1)$. The BIBD plan for $g = 9$ treatments in $b = 12$ blocks of size $k = 3$, given in Appendix C, is exactly the balanced lattice constructed above.

Balanced Lattice
($k + 1$ replicates)
is a BIBD

The (average) efficiency of a Square Lattice relative to an RCB is

$$E_{\text{SL:RCB}} = \frac{(k + 1)(r - 1)}{(k + 1)(r - 1) + r}.$$

This is the best possible efficiency for any resolvable design.

The *Rectangular Lattice* is closely related to the Square Lattice. Arrange the $g = k(k + 1)$ treatments in an $(k + 1) \times (k + 1)$ square with the diagonal blank, for example:

•	1	2	3
4	•	5	6
7	8	•	9
10	11	12	•

Rectangular
Lattice is subset
of a square

As with the Square Lattice, the first two replicates are formed from the rows and columns of this arrangement, ignoring the diagonal: (1, 2, 3), (4, 5, 6), (7, 8, 9), (10, 11, 12), (4, 7, 10), (1, 8, 11), (2, 5, 12), (3, 6, 9). Additional replicates are formed from the letters of orthogonal Latin Squares that satisfy the extra constraints that all the squares have the same diagonal and all letters appear on the diagonal; for example:

A	B	C	D		A	C	D	B
C	D	A	B		B	D	C	A
D	C	B	A		C	A	B	D
B	A	D	C		D	B	A	C

Rows, columns,
and Latin
Squares for a
Rectangular
Lattice

These squares are orthogonal and share the same diagonal containing all treatments. The next two replicates for this Rectangular Lattice design are thus (5, 9, 11), (1, 6, 10), (2, 4, 8), (3, 7, 12) and (6, 8, 12), (3, 4, 11), (1, 5, 7), (2, 9, 10).

The *Cubic Lattice* is a generalization the Square Lattice. In the Square Lattice, each treatment can be indexed by two subscripts i, j , with $1 \leq i \leq k$

Cubic Lattice for k^3 treatments in blocks of k

and $1 \leq j \leq k$. The subscript i indexes rows, and the subscript j indexes columns. The first row in the Square Lattice is all those treatments with $i = 1$. The second column is all those treatments with $j = 2$. The blocks of the first replicate of a Square Lattice are rows; that is, treatments are the same block if they have the same i . The blocks of the second replicate of the Square Lattice are columns; that is, treatments are in the same block if they have the same j .

Form blocks by keeping two subscripts constant

For the Cubic Lattice, we have $g = k^3$ treatments that we index with three subscripts i, j, l , with $1 \leq i \leq k$, $1 \leq j \leq k$, and $1 \leq l \leq k$. Each replicate of the Cubic Lattice will be k^2 blocks of size k . In the first replicate of a Cubic Lattice, treatments are grouped so that all treatments in a block have the same values of i and j . In the second replicate, treatments in the same block have the same values of i and l , and in the third replicate, treatments in the same block have the same values of j and l . For example, when $g = 8 = 2^3$, the cubic lattice will have four blocks of size two in each replicate. These blocks are as follows (using the ijl subscript to represent a treatment):

Replicate 1	Replicate 2	Replicate 3
(111, 112)	(111, 121)	(111, 211)
(121, 122)	(112, 122)	(112, 212)
(211, 212)	(211, 221)	(121, 221)
(221, 222)	(212, 222)	(122, 222)

Cubic Lattice designs can have 3, 6, 9, and so forth replicates by repeating this pattern.

Treatments adjusted for blocks

The intrablock Analysis of Variance for a Square, Cubic, or Rectangular Lattice is analogous to that for the BIBD; namely, treatments should be adjusted for blocks.

14.6 Alpha Designs

Alpha Designs are resolvable with $g = mk$

Alpha Designs allow us to construct resolvable incomplete block designs when the number of treatments g or block size k does not meet the strict requirements for one of the lattice designs. Alpha Designs require that the number of treatments be a multiple of the block size $g = mk$, so that there are m blocks per replication and $b = rm$ blocks in the complete design.

Three-step construction

We construct an Alpha Design in three steps. First we obtain the “generating array” for k , m , and r . This array has k rows and r columns. Next we expand each column of the generating array to m columns using a cyclic pattern to obtain an “intermediate array” with k rows and mr columns. Finally

we add m to the second row of the intermediate array, $2m$ to the third row, and so on. Columns of the final array are blocks.

Section C.4 has generating arrays for m from 5 to 15, k at least four but no more than the minimum of m and $100/m$, and r up to four. The major division is by m , so first find the full array for your value of m . We only need the first k rows and r columns of this full tabulated array.

Finding the
generating array

For example, suppose that we have $g = 20$ treatments and blocks of size $k = 4$, and we desire $r = 2$ replications. Then $m = 5$ and $b = 10$. The full generating array for $m = 5$ from Section C.4 is

1	1	1	1
1	2	5	3
1	3	4	5
1	4	3	2
1	5	2	4

We only need the first $k = 4$ rows and $r = 2$ columns, so our generating array is

1	1
1	2
1	3
1	4

Step two takes each column of the generating array and does cyclic substitution with $1, 2, \dots, m$, to get m columns. So, for our array, we get

Construct
intermediate
array

1	2	3	4	5	1	2	3	4	5
1	2	3	4	5	2	3	4	5	1
1	2	3	4	5	3	4	5	1	2
1	2	3	4	5	4	5	1	2	3

The first five columns are from the first column of the generating array, and the last five columns are from the last column of the generating array. This is the intermediate array.

Finally, we take the intermediate array and add $m = 5$ to the second row, $2m = 10$ to the third row, and $3m = 15$ to the last row, obtaining

Add multiples of
 m to rows

1	2	3	4	5	1	2	3	4	5
6	7	8	9	10	7	8	9	10	6
11	12	13	14	15	13	14	15	11	12
16	17	18	19	20	19	20	16	17	18

This is our final design, with columns being blocks and numbers indicating treatments.

The Alpha Designs constructed from the tables in Section C.4 are with a few exceptions the most efficient Alpha Designs possible. The average efficiencies for these Alpha Designs are very close to the theoretical upper bound for average efficiency of a resolvable design, namely

$$E_{\alpha:\text{RCB}} \leq \frac{(g-1)(r-1)}{(g-1)(r-1) + r(m-1)} .$$

14.7 Further Reading and Extensions

Incomplete block designs have been the subject of a great deal of research and theory; we have mentioned almost none of it. Two excellent sources for more theoretical discussions of incomplete blocks are John (1971) and John and Williams (1995). Among the topics relevant to this chapter, John (1971) describes recovery of interblock information for BIBD, PBIBD, and general incomplete block designs; existence and construction of BIBD's; classification, existence, and construction of PBIBD's; and efficiency. John and Williams (1995) is my basic reference for Cyclic Designs, Alpha Designs, and incomplete block efficiencies; and it has a good deal to say about row column designs, interblock information, and other topics as well.

Most of the designs described in this chapter are not recent. Many of these incomplete block designs were introduced by Frank Yates in the late 1930's, including BIBD's (Yates 1936a), Square Lattices (Yates 1936b), and Cubic Lattices (Yates 1939), as well other designs such as Lattice Squares (different from a Square Lattice, Yates 1940). PBIBD's first appear in Bose and Nair (1939). Alpha Designs are the relative newcomers, first appearing in Patterson and Williams (1976).

John and Williams (1995) provide a detailed discussion of the efficiencies of incomplete block designs, including a proof that the BIBD has the highest possible efficiency for equally replicated designs with equal block sizes. Section 3.3 of their book gives an expression for the efficiency of a cyclic design; Sections 2.8 and 4.10 give a variety of upper bounds for the efficiencies of blocked designs and resolvable designs. Chapter 12 of John (1971) and Chapter 1 of Bose, Clatworthy, and Shrikhande (1954) describe efficiency of PBIBD's.

Some experimental situations will not fit into any of the standard design categories. For example, different treatments may have different replication, or blocks may have different sizes. Computer software exists that will search

for “optimal” allocations of the treatments to units. *Optimal* can be defined in several ways; for example, you could choose to minimize the average variance for pairwise comparisons. See Silvey (1980) and Cook and Nachtsheim (1989).

14.8 Problems

Consider the following incomplete block experiment with nine treatments (A-I) in nine blocks of size three.

Exercise 14.1

Block								
1	2	3	4	5	6	7	8	9
C 54	B 35	A 48	G 46	D 61	C 52	A 54	B 45	A 31
H 56	G 36	G 42	H 56	E 61	I 53	H 59	I 46	B 28
D 53	D 40	E 43	I 59	F 54	E 48	F 62	F 47	C 25

- Identify the type of design.
- Analyze the data for differences between the treatments.

Chemical yield may be influenced by the temperature, pressure, and/or time in the reactor vessel. Each of these factors may be set at a high or a low level. Thus we have a 2^3 experiment. Unfortunately, the process feedstock is highly variable, so batch to batch differences in feedstock are expected; we must start with new feedstock every day. Furthermore, each batch of feedstock is only big enough for seven runs (experimental units). We have enough money for eight batches of feedstock. We decide to use a BIBD, with each of the eight factor-level combinations missing from one of the blocks.

Exercise 14.2

Give a skeleton ANOVA (source and degrees of freedom only), and describe an appropriate randomization scheme.

Briefly describe the following incomplete block designs (BIBD, or PBIBD with what associate classes, and so on).

Exercise 14.3

Block					Block					
	1	2	3	4		1	2	3	4	5
(a)	A	A	B	A	(b)	A	A	A	B	C
	B	C	C	B		B	B	C	D	D
	C	D	D	D		C	D	E	E	E
Block										
	1	2	3	4						
(c)	1	3	1	2						
	2	4	3	4						

Exercise 14.4

We wish to compare the average access times of five brands of half-height computer disk drives (denoted A through E). We would like to block on the computer in which they are used, but each computer will only hold four drives. Average access times and the design are given in the following table (data from Nelson 1993):

Computer				
1	2	3	4	5
A 35	A 41	B 40	A 32	A 40
B 42	B 45	C 42	C 33	B 38
C 31	D 32	D 33	D 35	C 35
D 30	E 40	E 39	E 36	E 37

Analyze these data and report your findings, including a description of the design.

Problem 14.1

Japanese beetles ate the Roma beans in our garden last year, so we ran an experiment this year to learn the best pesticide. We have six garden beds with beans, and the garden store has three different sprays that claim to keep the beetles off the beans. Sprays drift on the wind, so we cannot spray very small areas. We divide each garden bed into two plots and use a different spray on each plot. Below are the numbers of beetles per plot.

Bed					
1	2	3	4	5	6
19 A	9 A	25 B	9 A	26 A	13 B
21 B	16 C	30 C	11 B	33 C	18 C

Analyze these data to determine the effects of sprays. Which one should we use?

Problem 14.2

Milk can be strained through filter disks to remove dirt and debris. Filters are made by surface-bonding fiber webs to both sides of a disk. This experiment is concerned with how the construction of the filter affects the speed of milk flow through the filter.

We have a 2^4 factorial structure for the filters. The factors are fiber weight (normal or heavy), loft (thickness of the filter, normal or low), bonding solution on bottom surface (A or B), and bonding solution on top surface (A or B). Note the unfortunate fact that the “high” level of the second factor, loft, is low loft. Treatments 1 through 16 are the factor-level combinations in standard order.

These are speed tests, so we pour a measured amount of milk through the disk and record the filtration time as the response. We expect considerable

variation from farm to farm, so we block on farm. We also expect variation from milking to milking, so we want all measurements at one farm to be done at a single milking. However, only three filters can be satisfactorily used at a single milking. Thus we must use incomplete blocks of size three.

Sixteen farms were selected. At each farm there will be three strainings at one milking, with the milk strained first with one filter, then a second, then a third. Each treatment will be used three times in the design: once as a first filter, once as second, and once as third. The treatments and responses for the experiment are given below (data from Connor 1958):

Treatments and Responses							
Farm	Filtration time						
	First	Second	Third				
1	10	451	7	457	16	343	
2	11	260	8	418	13	320	
3	12	464	5	317	14	315	
4	9	306	6	462	15	291	
5	13	381	4	597	6	491	
6	14	362	1	325	7	449	
7	15	292	2	402	8	576	
8	16	431	3	477	5	394	
9	7	329	9	261	4	430	
10	8	389	10	413	1	272	
11	5	368	11	244	2	447	
12	6	398	12	517	3	354	
13	2	490	16	311	9	278	
14	3	467	13	429	10	486	
15	4	735	14	642	11	474	
16	1	402	15	380	12	589	

What type of design is this? Analyze the data and report your findings on the influence of the treatment factors on straining time.

The State Board of Education has adopted basic skills tests for high school graduation. One of these is a writing test. The student writing samples are graded by professional graders, and the board is taking some care to be sure that the graders are grading to the same standard. We examine grader differences with the following experiment. There are 25 graders. We select 30 writing samples at random; each writing sample will be graded by five graders. Thus each grader will grade six samples, and each pair of graders will have a test in common.

Problem 14.3

Exam	Grader					Score					Exam	Grader					Score				
1	1	2	3	4	5	60	59	51	64	53	16	1	9	12	20	23	61	67	69	68	65
2	6	7	8	9	10	64	69	63	63	71	17	2	10	13	16	24	78	75	76	75	72
3	11	12	13	14	15	84	85	86	85	83	18	3	6	14	17	25	67	72	72	75	76
4	16	17	18	19	20	72	76	77	74	77	19	4	7	15	18	21	84	81	76	79	77
5	21	22	23	24	25	65	73	70	71	70	20	5	8	11	19	22	81	84	85	84	81
6	1	6	11	16	21	52	54	62	54	55	21	1	8	15	17	24	70	65	61	66	66
7	2	7	12	17	22	56	51	52	57	51	22	2	9	11	18	25	84	82	86	85	86
8	3	8	13	18	23	55	60	59	60	61	23	3	10	12	19	21	72	85	77	82	79
9	4	9	14	19	24	88	76	77	77	74	24	4	6	13	20	22	85	75	78	82	83
10	5	10	15	20	25	65	68	72	74	77	25	5	7	14	16	23	58	64	58	57	58
11	1	10	14	18	22	79	77	77	77	79	26	1	7	13	19	25	66	71	73	70	70
12	2	6	15	19	23	70	66	63	62	66	27	2	8	14	20	21	73	67	63	70	66
13	3	7	11	20	24	48	49	51	48	50	28	3	9	15	16	22	58	70	69	61	71
14	4	8	12	16	25	75	64	75	68	65	29	4	10	11	17	23	95	84	88	88	87
15	5	9	13	17	21	79	77	81	79	83	30	5	6	12	18	24	47	47	51	49	56

Analyze these data to determine if graders differ, and if so, how. Be sure to describe the design.

Problem 14.4

Thirty consumers are asked to rate the softness of clothes washed by ten different detergents, but each consumer rates only four different detergents. The design and responses are given below:

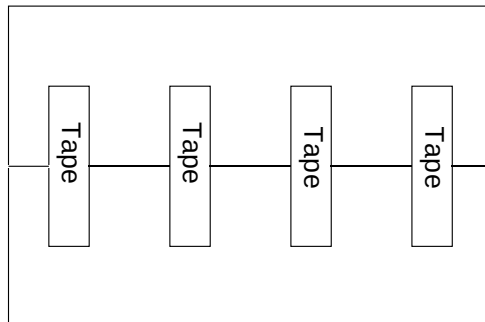
	Trts				Softness					Trts				Softness			
1	A	B	C	D	37	23	37	41	16	A	B	C	D	52	41	45	48
2	A	B	E	F	35	32	39	37	17	A	B	E	F	46	42	45	42
3	A	C	G	H	39	45	39	41	18	A	C	G	H	44	43	41	36
4	A	D	I	J	44	42	46	44	19	A	D	I	J	32	42	36	29
5	A	E	G	I	44	44	45	50	20	A	E	G	I	43	42	44	44
6	A	F	H	J	55	45	53	49	21	A	F	H	J	46	41	43	45
7	B	C	F	I	47	50	48	52	22	B	C	F	I	43	51	40	42
8	B	D	G	J	37	42	40	37	23	B	D	G	J	38	37	36	34
9	B	E	H	J	32	34	39	29	24	B	E	H	J	40	49	43	44
10	B	G	H	I	36	41	39	43	25	B	G	H	I	23	20	27	29
11	C	E	I	J	45	44	40	36	26	C	E	I	J	46	49	48	43
12	C	F	G	J	42	38	39	39	27	C	F	G	J	48	43	48	41
13	C	D	E	H	47	48	46	47	28	C	D	E	H	35	35	31	26
14	D	E	F	G	43	47	48	41	29	D	E	F	G	45	47	47	42
15	D	F	H	I	39	32	32	31	30	D	F	H	I	43	39	38	39

Analyze these data for treatment effects and report your findings.

Briefly describe the experimental design you would choose for each of the following situations, and why.

Problem 14.5

- (a) Competition cuts tree growth rate, so we wish to study the effects on tree growth of using four herbicides on the competition. There are many study sites available, but each site is only large enough for three plots. Resources are available for 24 plots (that is, eight sites with three plots per site). Large site differences are expected.
- (b) We use 2-inch wide tape to seal moving cartons, and we want to find the brand that seals best. The principal problem is not the tape breaking, but the tape pulling away from the cardboard. Unfortunately, there is considerable variation from carton to carton in the ability of any tape to adhere to the cardboard. There are four brands of tape available. The test is to seal a box bottom with four strips of tape of one or more types, place the carton so that only the edges are supported, drop 50 pounds of old *National Geographic* into the carton from a height of one foot, and then measure the length of tape that pulled away from the cardboard. There is a general tendency for tape to pull away more in the center of the carton than near its ends. Our cheap boss has given us only sixteen boxes to ruin in this destructive fashion before deciding on a tape. Tape placement on the bottom looks like this:



- (c) Three treatments are being studied for the rehabilitation of acidified lakes. Unfortunately, there is tremendous lake to lake variability, and we only have six lakes on which we are allowed to experiment. We may treat each lake as a whole, or we may split each lake in two using a plastic “curtain” and treat the halves separately. Sadly, the technology does not allow us to split each lake into three.
- (d) A retail bookstore has two checkouts, and thus two checkout advertising displays. These displays are important for enticing impulse purchases, so the bookstore would like to know which of the four types of

displays available will lead to the most sales. The displays will be left up for one week, because it is expensive to change displays and you really need a full week to get sufficient volume of sales and overcome day-of-week effects; there are, however, week to week differences in sales. The store wishes to complete the comparison in at most 8 and preferably fewer weeks.

- (e) We wish to compare four “dog collars.” The thought is that some collars will lead to faster obedience than others. The response we measure will be the time it takes a dog to complete a walking course with lots of potential distractions. We have 24 dogs that can be used, and we expect large dog to dog variability. Dogs can be used more than once, but if they are used more than once there should be at least 1 week between trials. Our experiment should be completed in less than 3 weeks, so no dog could possibly be used more than three times.

Problem 14.6

For each of the following, describe the experimental design that was used, and give a skeleton ANOVA.

- (a) Plant breeders wish to study six varieties of corn. They have 24 plots available, four in each of six locations. The varieties are assigned to location as follows (there is random assignment of varieties to plot within location):

Locations					
1	2	3	4	5	6
A	B	A	A	B	A
B	C	C	B	C	C
D	E	D	D	E	D
E	F	F	E	F	F

- (b) We wish to study gender bias in paper grading. We have 12 “lower” level papers and 12 “advanced” level papers. There are four paid graders who do not know the students or their names. Each paper is submitted for grading exactly once (that is, no paper is graded by more than one grader). We examine gender bias by the name put on the paper: either a male first name, a female first name, or just initials. The twelve lower-level papers are assigned at random to the combinations of grader and name gender, as are the advanced-level papers. The response we measure is the grade given (on a 0-100 scale).
- (c) Song bird abundance can be measured by sending trained observers to a site to listen for the calls of the birds and make counts. Consider an experiment on the effects of three different forest harvesting techniques

on bird abundance. There are six forests and two observers, and there will be two harvests in each of the six forests. The harvest techniques were assigned in the following way:

Observer	Forest					
	1	2	3	4	5	6
1	A	C	B	B	A	C
2	C	A	A	C	B	B

- (d) Wafer board is a manufactured wood product made from wood chips. One potential problem is warping. Consider an experiment where we compare three kinds of glue and two curing methods. All six combinations are used four times, once for each of four different batches of wood chips. The response is the amount of warping.

When recovering interblock information in a BIBD, we take the weighted average of intra- and interblock estimates

$$\bar{\zeta} = \lambda \hat{\zeta} + (1 - \lambda) \tilde{\zeta} .$$

Suppose that $\sigma^2 = \sigma_\beta^2 = 1$, $g = 8$, $k = 7$, and $b = 8$. Find the mean and standard deviation of $1/\lambda$. Do you feel that λ is well determined?

Question 14.1